

The background of the slide features a close-up of a vibrant green leaf, likely from a citrus tree, showing detailed vein patterns. In the lower-left foreground, a clear glass filled with water is visible, with light reflecting off its surface. The entire scene is overlaid with a semi-transparent dark blue-grey filter to ensure the white text is legible.

AGRICULTURE LEAF DISEASE CLASSIFICATION USING DEEP LEARNING

GUIDED BY : MRS.
S.SANGEETHA

Submitted by

RAVEEN D (960121243038)

DEEPAK M (960121243012)

RACHEL D (960121243035)

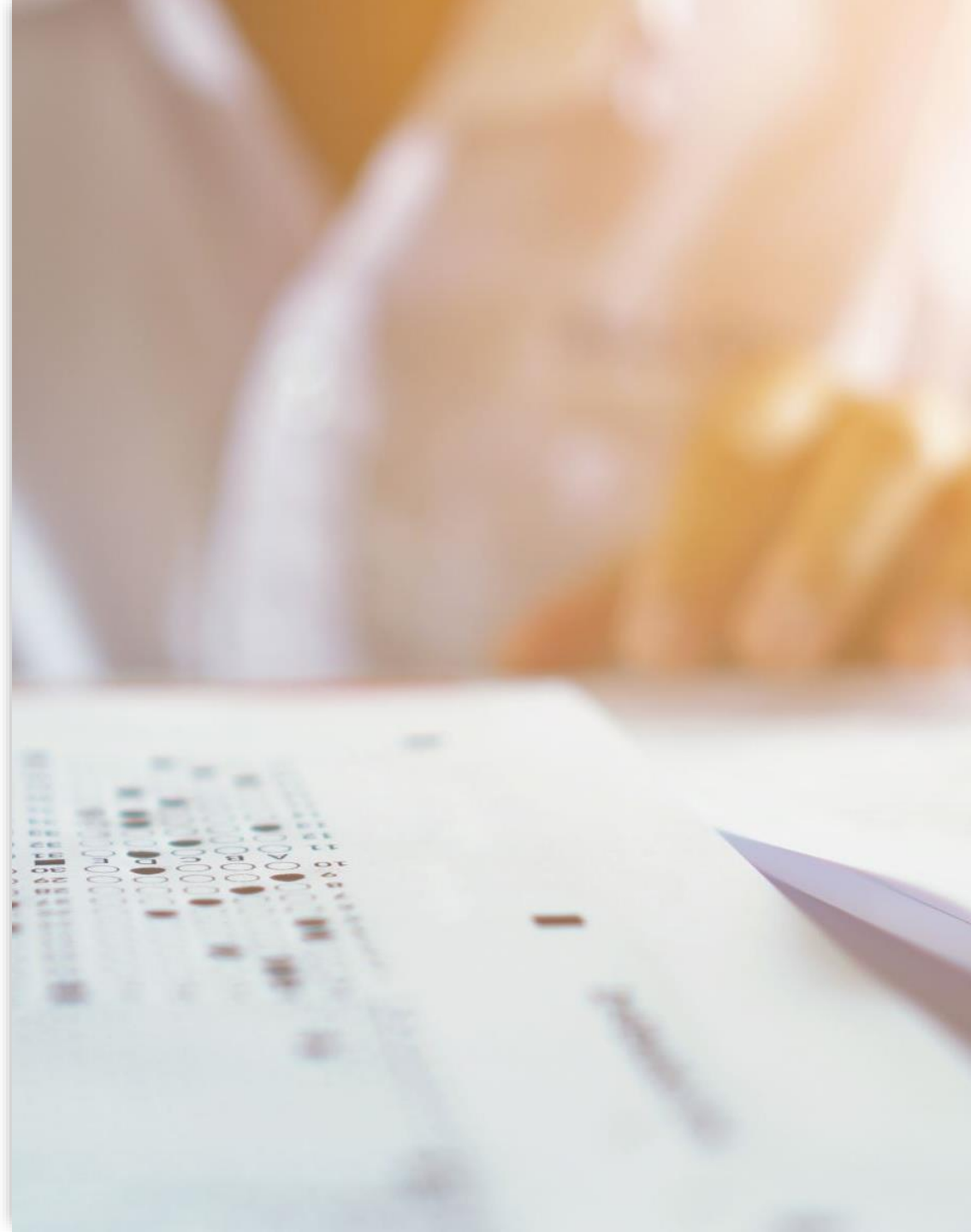
ANUSHA S (960121243006)

Abstract

Agriculture leaf disease classification using deep-learning focuses on the detection of plant diseases using deep learning techniques, specifically leveraging the MobileNetV2 architecture. The Plant Village - Agricultural Leaf Disease Dataset, which includes a large collection of leaf images from 42 disease classes, is used for training, testing, and validating the model. The dataset comprises 28,830 images in both the training and test sets, with an additional 11,527 images in the validation set. The images are categorized into different classes representing various plant diseases or healthy leaves. The core of the detection system is MobileNetV2, a lightweight and efficient deep convolutional neural network (CNN) architecture developed by Google. MobileNetV2 is designed for mobile and embedded vision applications, balancing high accuracy with low computational cost. It utilizes depthwise separable convolutions, significantly reducing the number of parameters and operations compared to traditional CNNs, making it suitable for resource-constrained devices. Moreover, Inverted Residuals and Linear Bottlenecks allow the model to maintain performance with fewer resources, ensuring efficient feature extraction for image classification tasks. Pre-trained on the ImageNet dataset, MobileNetV2 is fine-tuned for plant disease detection, including conditions such as Apple Scab, Leaf Blast, and Powdery Mildew. The model can be deployed on mobile devices and IoT systems for real-time plant disease classification and monitoring. This system aims to automate plant disease detection, providing an effective tool for agricultural monitoring, ensuring timely intervention, and reducing the spread of plant diseases.

Problem Statement

- Manual disease detection is time-consuming and inaccurate; AI automates this process, enabling faster and more accurate diagnoses.





Overview of Dataset

- **Dataset Name:**
 - Plant Village - Agricultural Leaf Disease Dataset
- **Description:**
 - The dataset is a large collection of leaf images for various crop plants. Each image is categorized into a specific disease class or marked as healthy.
- **Dataset Composition:**
 - Training Set:
 - Total Images: 28,830
 - Number of Classes: 42
 - Test Set:
 - Total Images: 28,830
 - Number of Classes: 42
 - Validation Set:
 - Total Images: 11,527
 - Number of Classes: 42

Classes in Dataset

Class	Disease Name	Class	Disease Name	Class	Disease Name	Class	Disease Name
1	Apple_scab	13	Grape Esca	25	Rice Brown Spot	37	Tomato Septoria Leaf Spot
2	Apple Black rot	14	Grape Leaf Blight	26	Rice Healthy	38	Tomato Spider Mites
3	Cedar apple rust	15	Grape Healthy	27	Rice Hispa	39	Tomato Target Spot
4	Apple healthy	16	Orange Huanglongbing	28	Rice Leaf Blast	40	Tomato Yellow Leaf Curl Virus
5	Blueberry healthy	17	Peach Bacterial Spot	29	Soybean Healthy	41	Tomato Mosaic Virus
6	Cherry Powdery mildew	18	Peach Healthy	30	Squash Powdery Mildew	42	Tomato Healthy
7	Cherry healthy	19	Pepper Bacterial Spot	31	Strawberry Leaf Scorch		
8	Corn Cercospora leaf spot	20	Pepper Healthy	32	Strawberry Healthy		
9	Corn Common rust	21	Potato Early Blight	33	Tomato Bacterial Spot		
10	Corn Northern Leaf Blight	22	Potato Late Blight	34	Tomato Early Blight		
11	Corn healthy	23	Potato Healthy	35	Tomato Late Blight		
12	Grape Black rot	24	Raspberry Healthy	36	Tomato Leaf Mold		

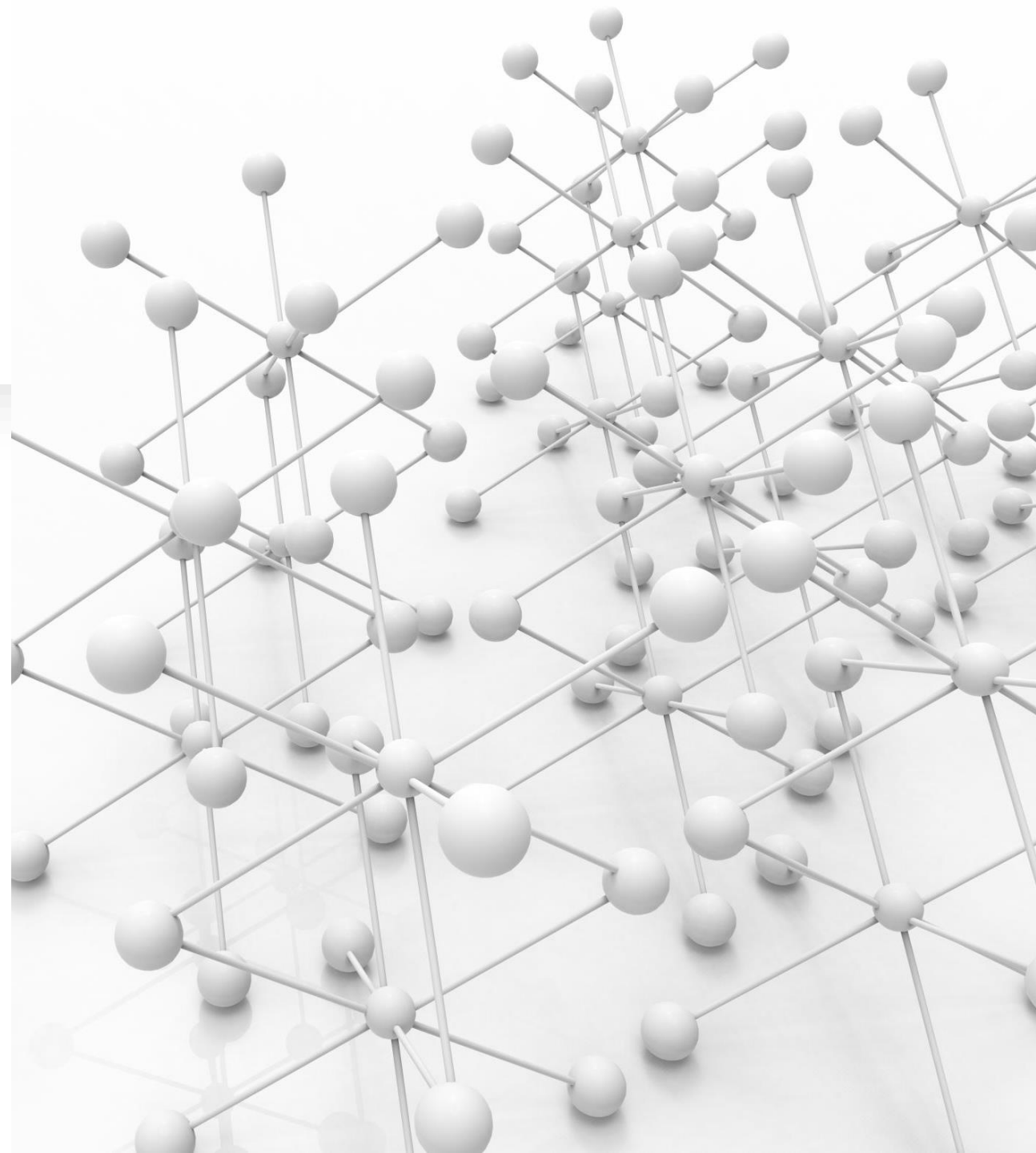


Disease Detection Process

- Images of plant leaves are fed into MobilenetV2 for classification.

MobileNetV2Model

- MobileNetV2 is a lightweight deep convolutional neural network (CNN) architecture developed by Google.
- Designed for mobile and embedded vision applications, it balances high accuracy with low computational cost and memory usage.
- Uses depthwise separable convolutions, significantly reducing the number of parameters and operations compared to standard convolutions.
- Introduces "Inverted Residuals" and "Linear Bottlenecks", which help the model learn efficient representations while maintaining performance, even with fewer resources.
- Efficient at handling image classification tasks on resource-constrained devices, such as smartphones and IoT devices.
- Pre-trained on the ImageNet dataset, enabling transfer learning for custom applications like plant disease detection, medical imaging, and real-time object classification.



Classes Detected by MobileNetV2

- MobileNetV2 is an efficient deep learning model suitable for plant disease detection tasks, including diseases like Apple Scab, Leaf Blast, Powdery Mildew, and more.



Building the Custom MobileNetV2Model for Plant Disease Classification

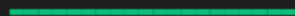
```
base_model = MobileNetV2(weights='imagenet', include_top=False, input_shape=(224, 224, 3))
base_model.trainable = False # Freeze base model

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(1024, activation='relu')(x)
x = Dense(512, activation='relu')(x)
output_layer = Dense(NUM_CLASSES, activation='softmax')(x)

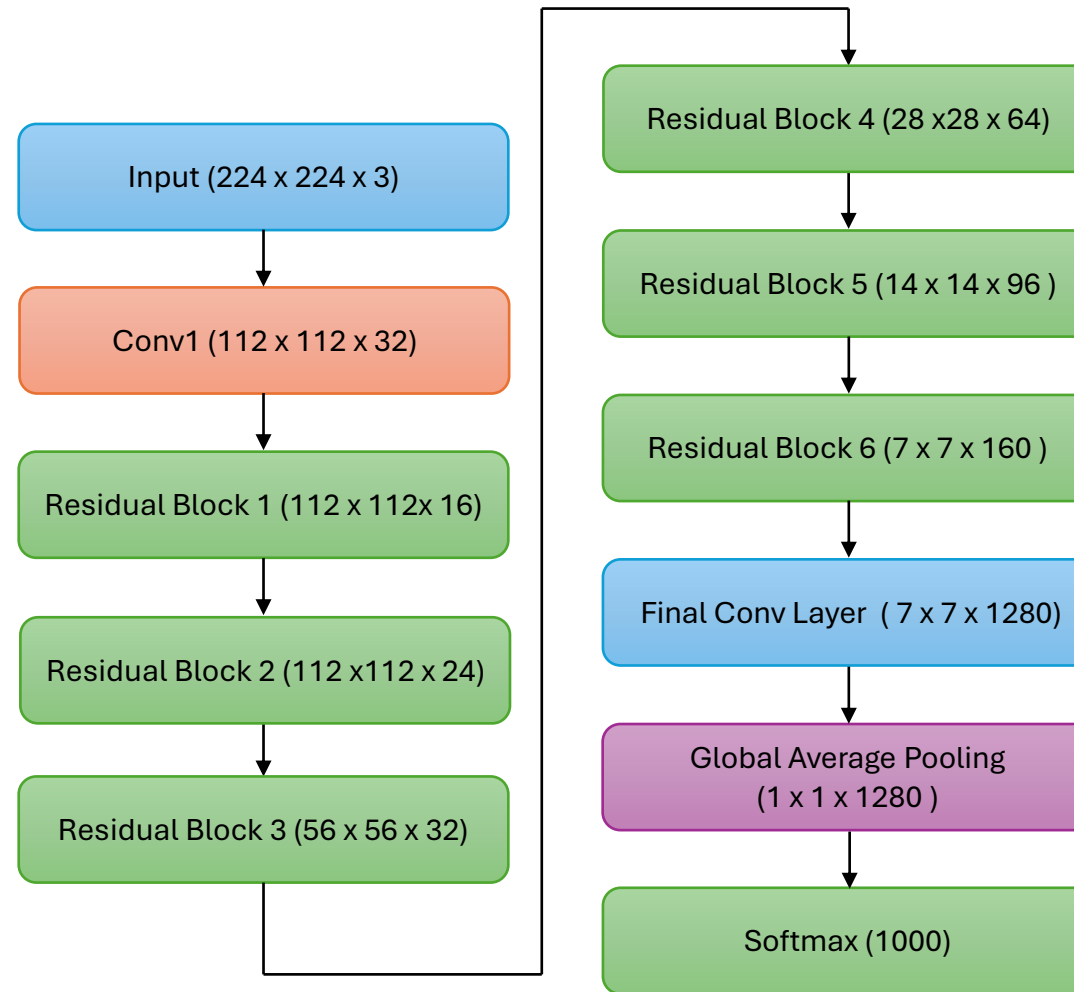
model = Model(inputs=base_model.input, outputs=output_layer)
```

[7]

Python

... Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_9406464/9406464  1s 0us/step

Model Architecture



Training and Evaluation Process



The model is trained over multiple epochs to learn patterns from the training data.



After each epoch, training accuracy and loss are recorded to monitor learning progress.



The model is also evaluated on a separate test dataset to assess generalization performance.



Accuracy and loss values are stored after each epoch to visualize performance trends.



This helps in identifying overfitting, underfitting, or the need for more epochs.

Epoch-wise Training Progress

901/901  652s 700ms/step - accuracy: 0.5870 - loss: 1.5902

901/901  125s 133ms/step - accuracy: 0.8750 - loss: 0.4556

✓ Epoch 1: Train Acc = 0.7292, Test Acc = 0.8373

🔄 Training Epoch 2/10

901/901  560s 617ms/step - accuracy: 0.8526 - loss: 0.4672

901/901  117s 130ms/step - accuracy: 0.8733 - loss: 0.4147

✓ Epoch 2: Train Acc = 0.8596, Test Acc = 0.8647

🔄 Training Epoch 3/10

901/901  563s 620ms/step - accuracy: 0.8736 - loss: 0.3749

901/901  112s 124ms/step - accuracy: 0.9108 - loss: 0.2783

✓ Epoch 3: Train Acc = 0.8791, Test Acc = 0.8994

🔄 Training Epoch 4/10

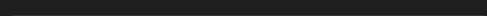
901/901  565s 622ms/step - accuracy: 0.8917 - loss: 0.3255

901/901  123s 137ms/step - accuracy: 0.9184 - loss: 0.2545

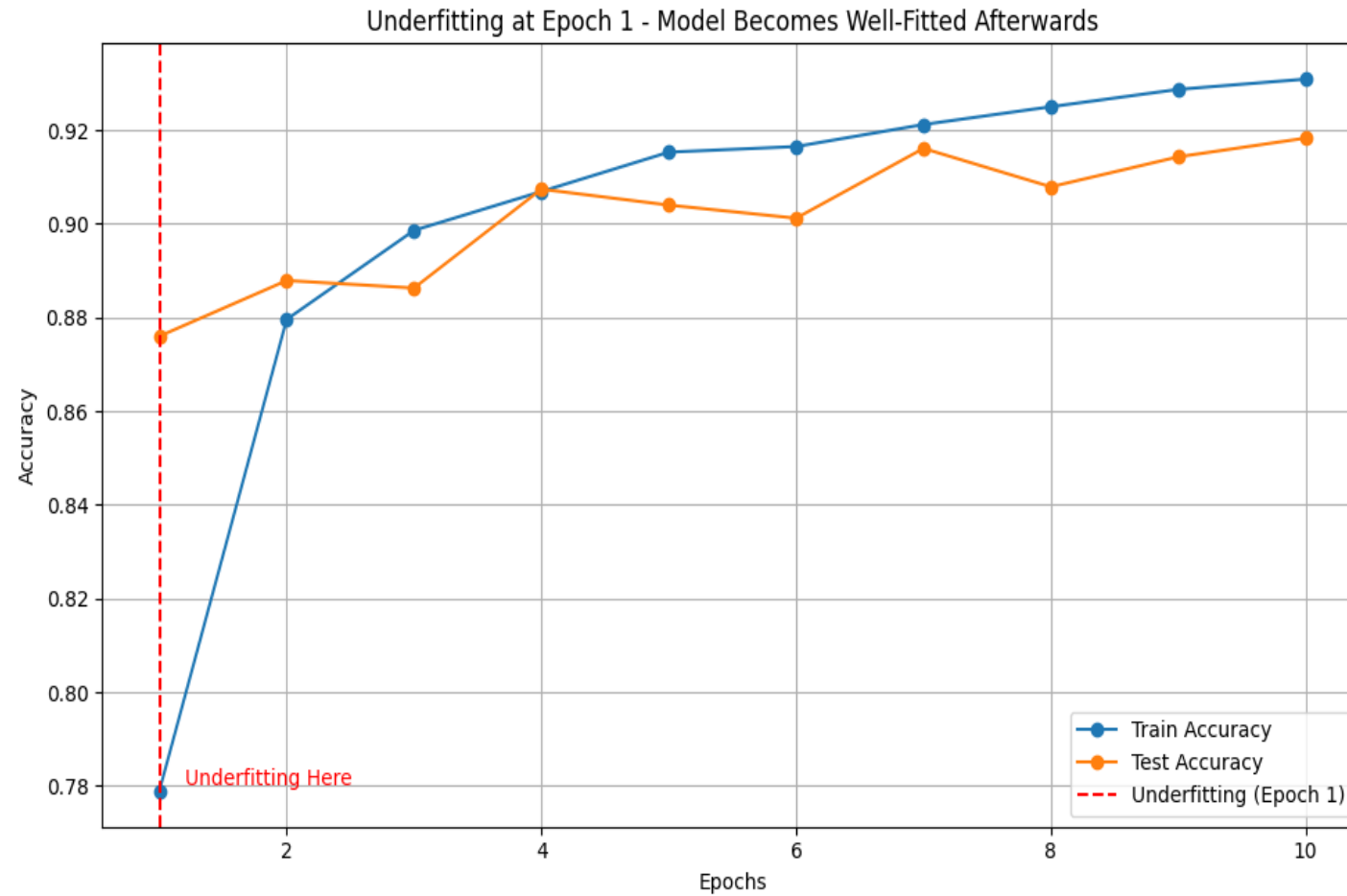
✓ Epoch 4: Train Acc = 0.8930, Test Acc = 0.9014

🔄 Training Epoch 5/10

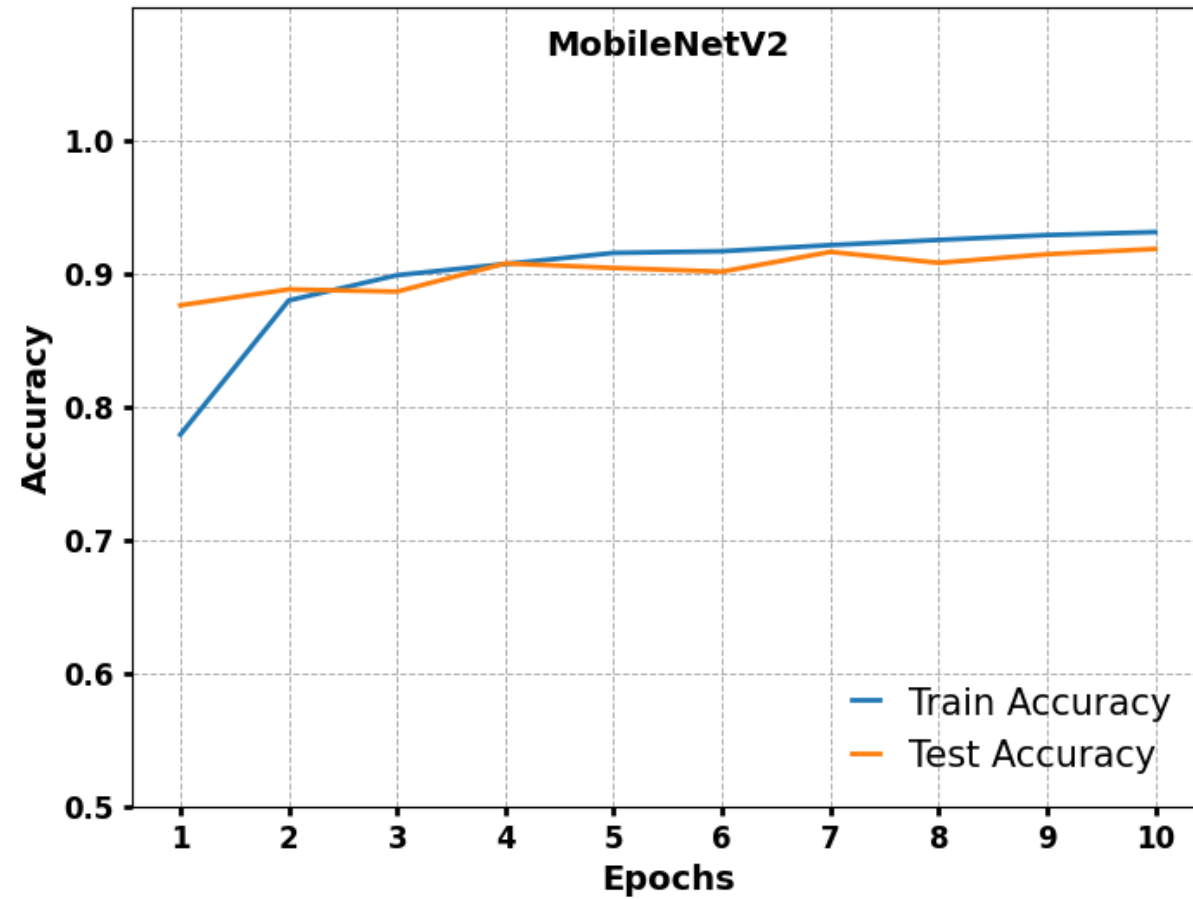
901/901  564s 622ms/step - accuracy: 0.8995 - loss: 0.2993

901/901  113s 125ms/step - accuracy: 0.9145 - loss: 0.2620

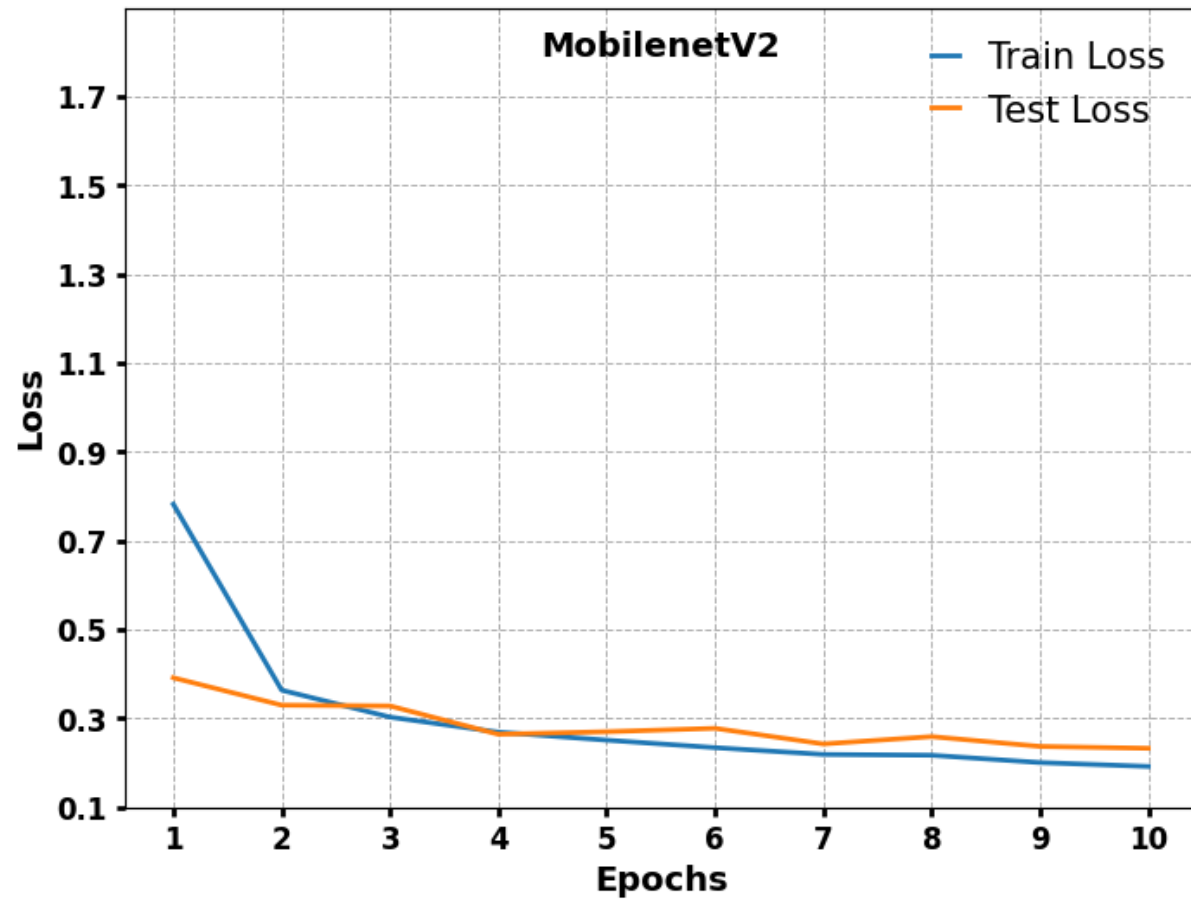
Underfitting



Model Train and Test Accuracy

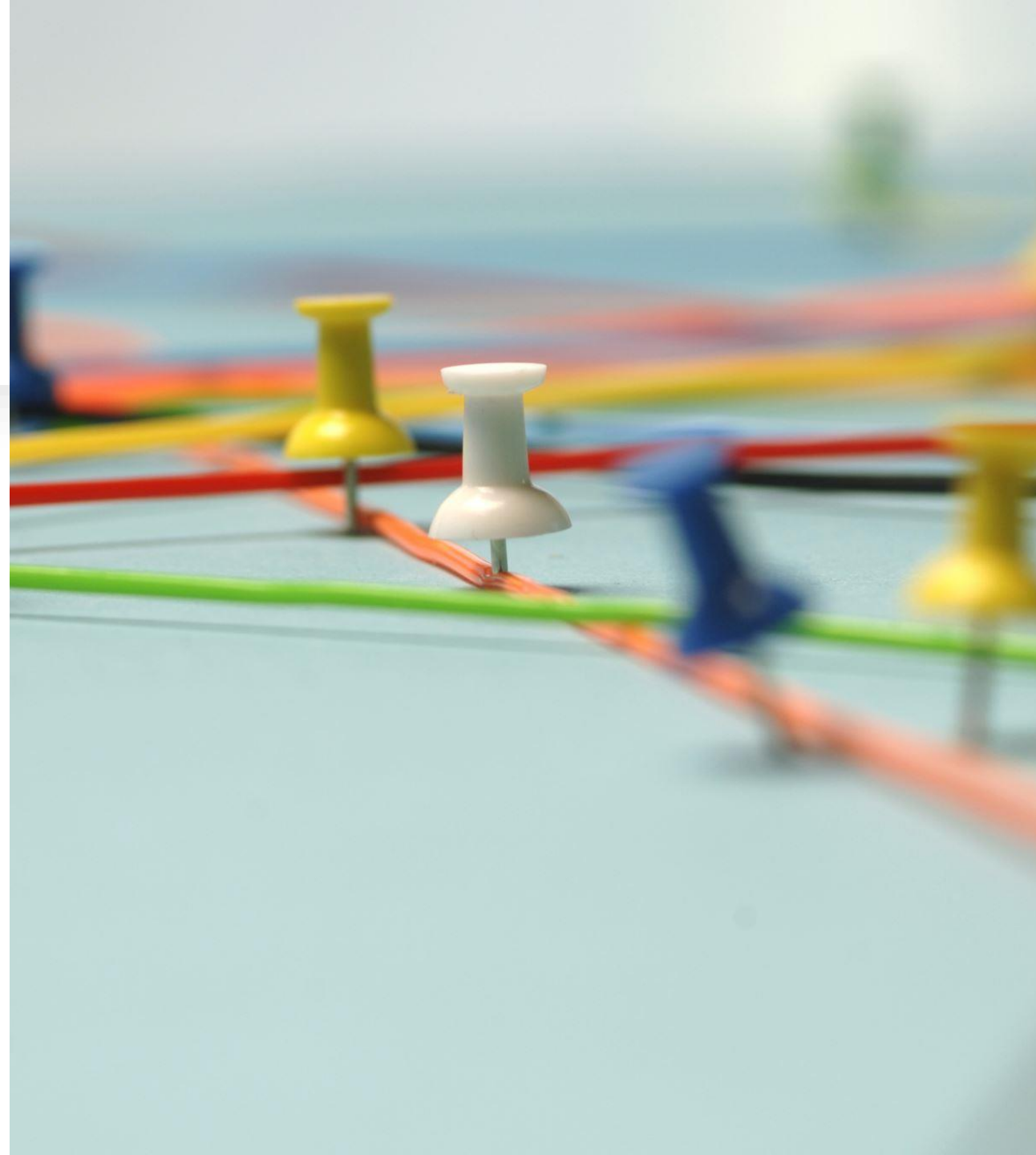


Model Train and Test Loss



Model Accuracy

Model	Train Accuracy (%)	Test Accuracy (%)	Validation Accuracy (%)
MobileNetV2	93.45	91.83	92.45

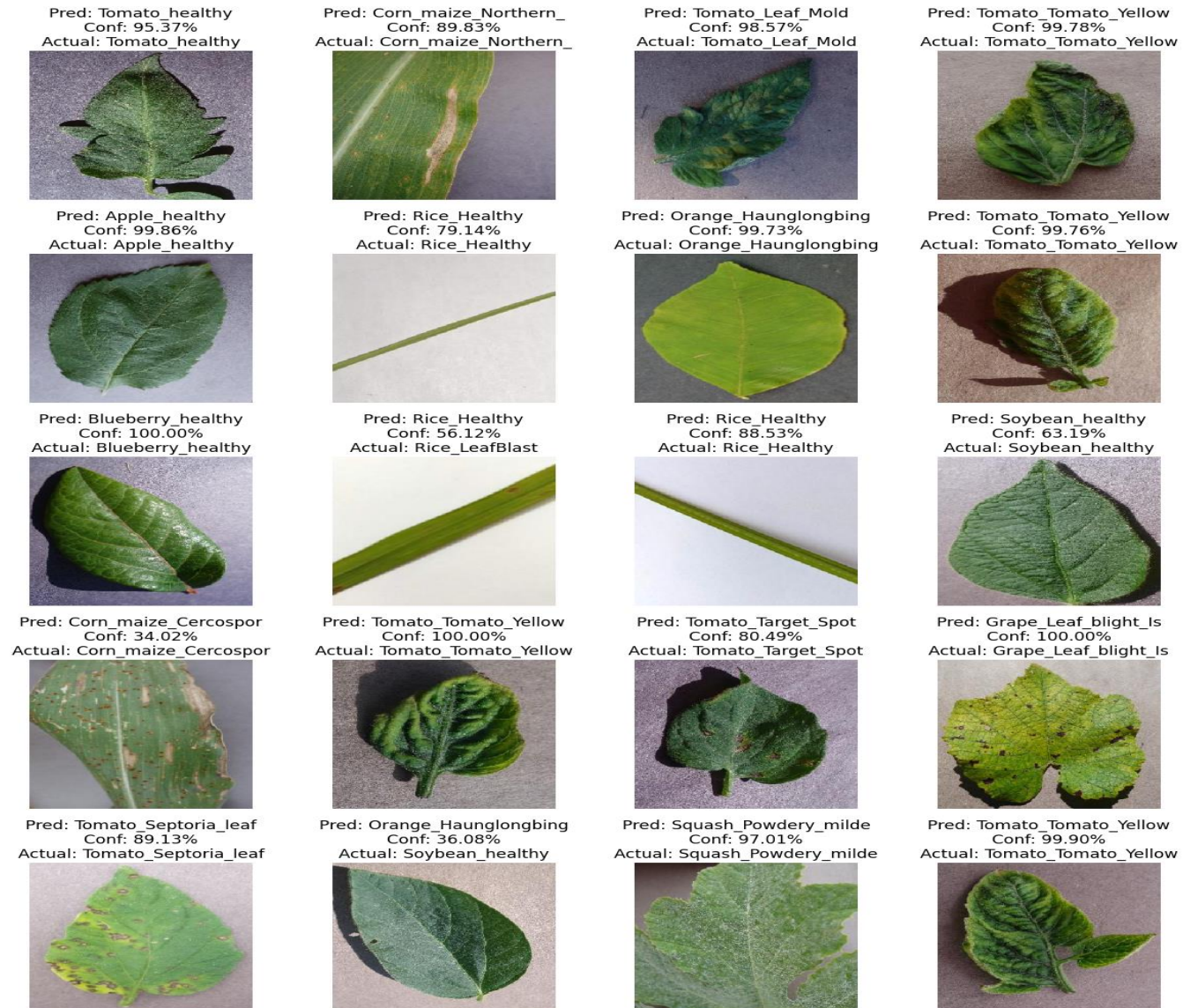


Testing the Model

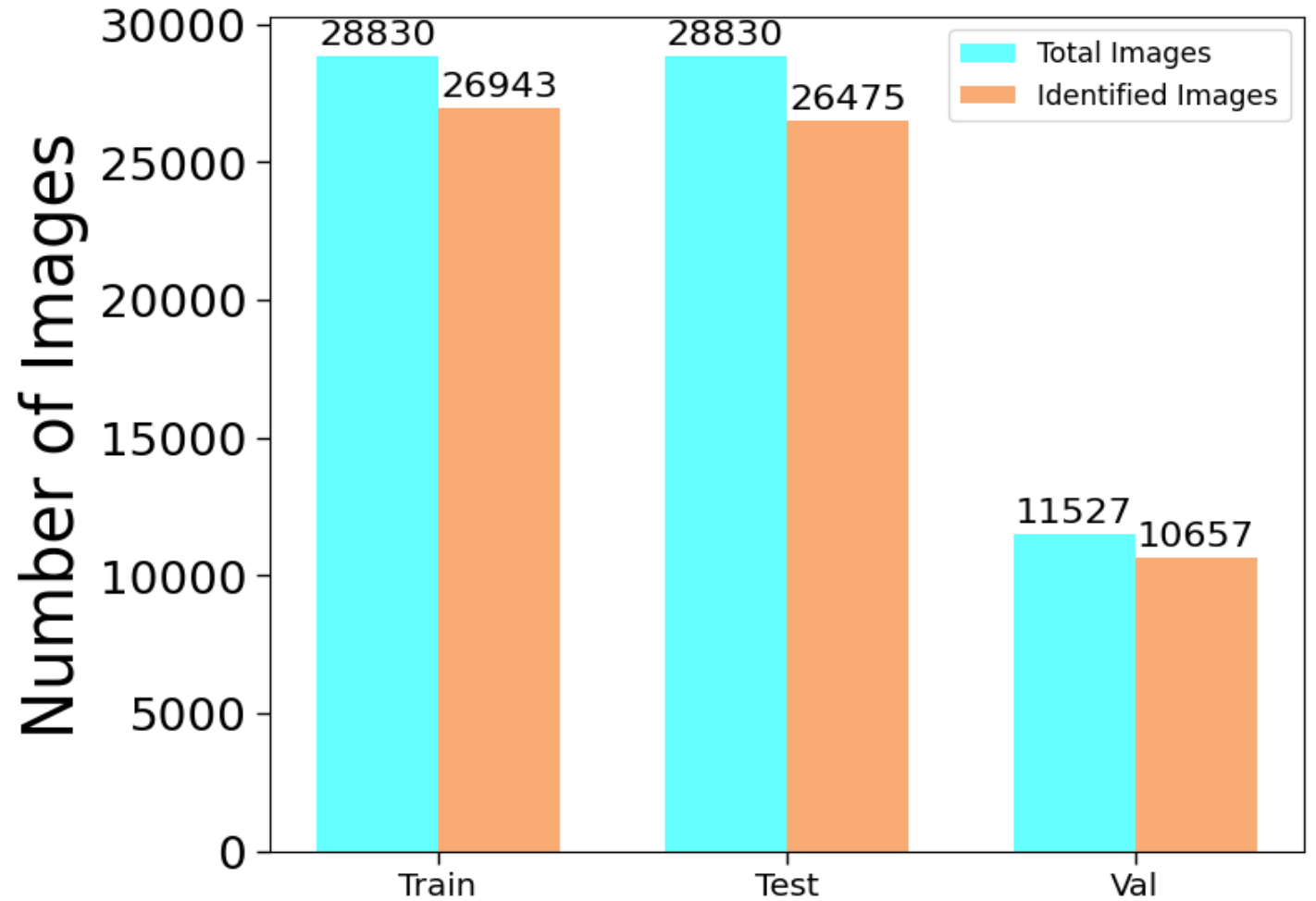
- Test model with unseen images to from validation data ensure generalization.



Predicted vs Actual

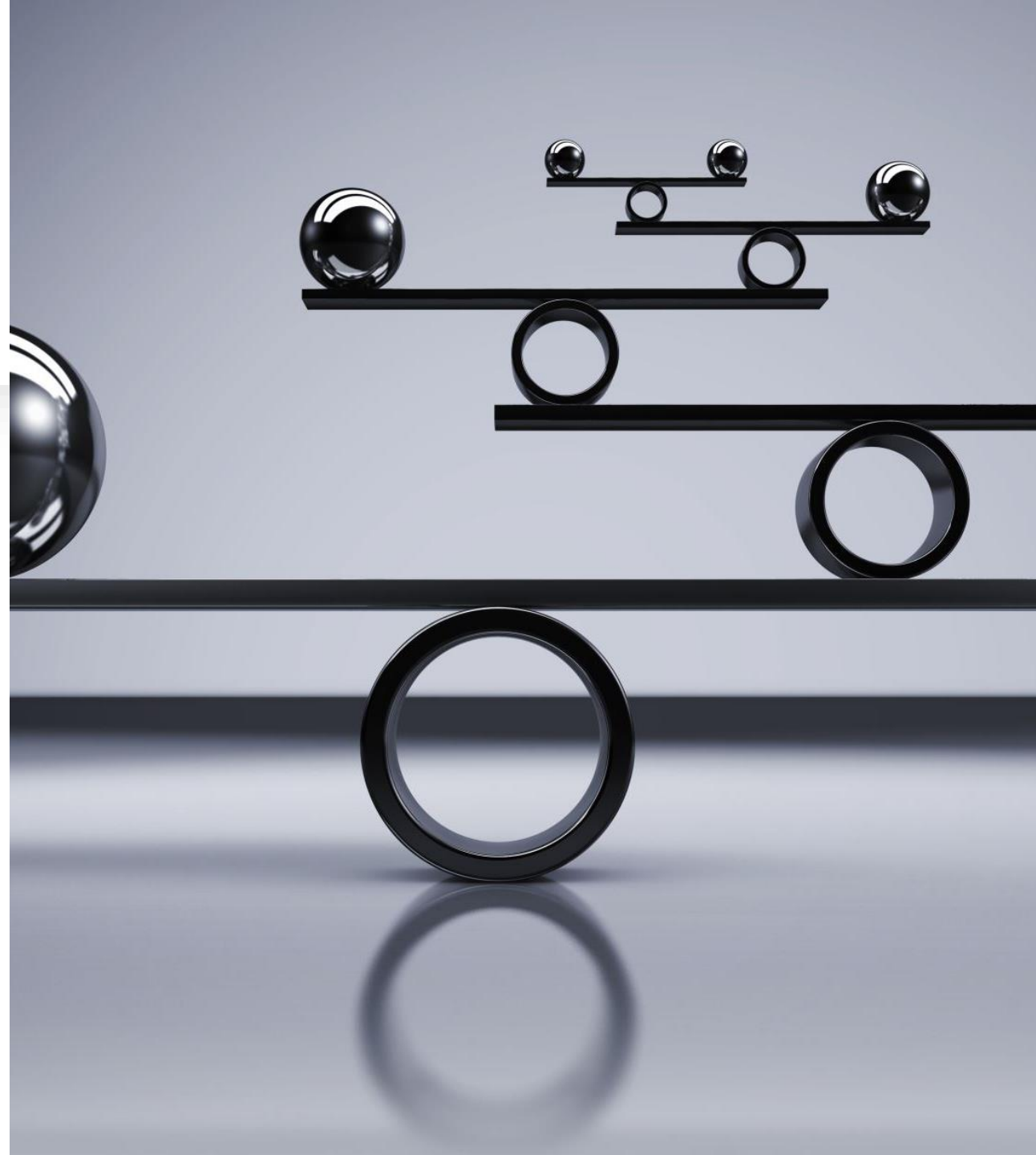


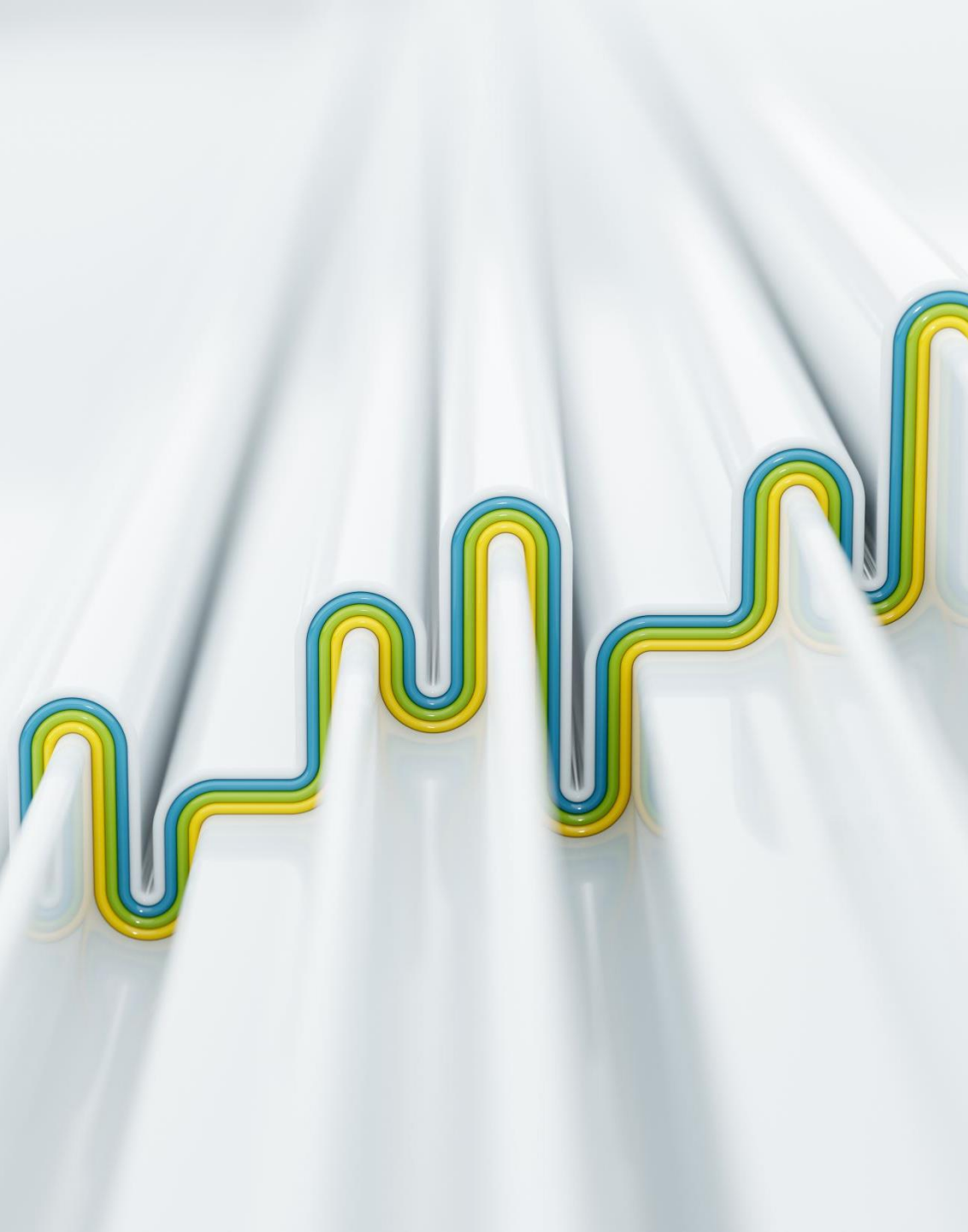
Model
Accuracy:
Correct Class
vs Total
Images Graph



Challenges in Disease Detection

- Challenges include class imbalance, overfitting, and dataset size.

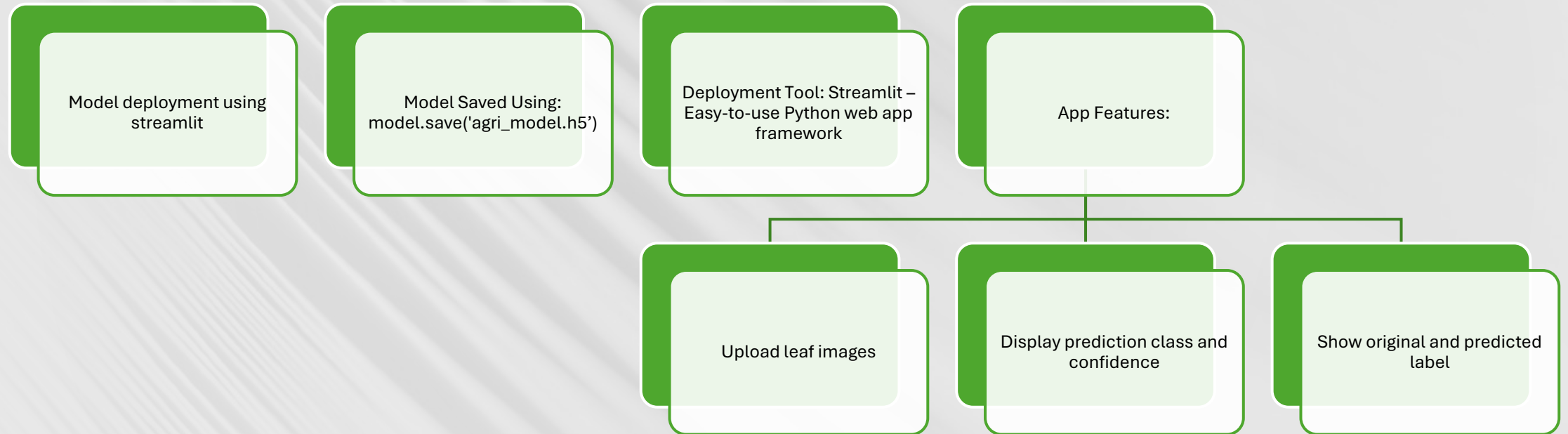




Solutions and Improvements

- Solutions involve data augmentation, dropout layers, and larger datasets.

Deployment



Deployment Results



Leaf Disease Detector

Upload a leaf image to detect the disease and get management tips.

Upload a leaf image...



Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files



IMG-20250513-WA0003.jpg 19.6KB



The use_column_width parameter has been deprecated and will be removed in a future release. Please utilize the use_container_width parameter instead.



Uploaded Image



Prediction Result

Detected Disease: Apple__Black_rot

Confidence: 68.23%

Severity: Moderate

Disease Management Tip:

Prune infected areas, use resistant varieties, and apply copper-based fungicides.

Role of AI in Agriculture

- AI in agriculture enables early disease detection, reducing chemical reliance and improving crop management.



Use Cases in Agriculture

- AI aids precision farming and minimizes pesticide use, improving disease detection accuracy.





Impact on Farmers

- Timely treatments prevent losses, benefiting farmers and increasing crop yield.

Future Scope

Future improvements: real-time monitoring, edge devices, better model generalization.

Conclusion

- Summary: AI's role in enhancing agriculture through automated disease detection.



References

Sources: Kaggle, PlantVillage,
TensorFlow Docs, Research Articles.





Thank You / Q&A

- Thank you for your attention! Any questions?